

Defects in Semiconductors

Thermodynamics, Electronic Consequences, and Characterization:
From Atomic Structure to Device Performance

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Thermodynamics of a Perfect Crystal

- The perfect crystal is an infinite idealized lattice of atoms that exhibits perfect translational symmetry, i.e, every point in the lattice is geometrically identical to every equivalent point. Though in reality, it is impossible to have a perfect crystal at any temperature $T > 0K$, due to Thermodynamics.
- The formation of any defect costs energy, known as the formation enthalpy (H). However, the creation of defects also introduces disorder into the crystal, which significantly increases the crystal's entropy (S).
- The system's goal is to minimize the Gibbs Free Energy ($G = H - TS$). As the temperature rises, the $-TS$ term becomes more significant, making it favourable for the crystal to intentionally create defects. Thus, defects are not just "mistakes" during growth; they are an intrinsic, thermodynamically required feature of any real material in equilibrium.
- These defects can be either detrimental or functional.

Point Defects

- Point defects, or zero-dimensional defects, are localized to a single lattice site or a position in between sites, and are classified into intrinsic (involving only the host atoms) or extrinsic (involving a foreign impurity atom).
- The Kröger-Vink notation (M_S^C) is used as a standard notation and describes the defect species (M), the lattice site it occupies (S), and its relative charge (C) with respect to the "perfect" lattice.
- Point defects are fundamentally electronic entities, not merely structural voids. The dangling bonds created by these defects introduce localized states that often sit inside the forbidden band gap.
- The charge of a defect is dynamic and depends on the position of the Fermi Level (E_F). Thus, the same defect can be neutral, positive, or negative depending on the material's doping. Charge Transition Levels (CTL) is the thermodynamic tipping point where the defect transitions from one stable charge state to the other.

Extended Defects - 1D, 2D, 3D

- 1D - Line defects, or dislocations, represent extended distortions in the crystal lattice. Edge dislocations correspond to an extra half-plane of atoms, with a Burgers vector perpendicular to the dislocation line. Screw dislocations produce a helical lattice twist, with Burgers vector parallel to the dislocation.
- 2D - Planar defects represent a breakdown of periodicity across a plane. The most prominent examples are grain boundaries, which are the interfaces where two crystals of different orientations meet. Stacking faults are another common 2D defect, representing an error in the stacking sequence of crystal planes.
- 3D - Volume Defects are bulk defects such as voids which are essentially large clusters of vacancies, or precipitates (for example, clusters of oxygen atoms in Czochralski-grown silicon or metallic silicide precipitates).

Diffusion of Defects

- The process of diffusion allows both host atoms and impurities to move through the lattice. For a species to be able to diffuse, it needs to cross the activation energy for migration E_m . The total diffusivity of a species is often a product of both the formation energy E_f and migration energy E_m .
- There are three primary mechanisms through which defects diffuse:
 - ① Vacancy Mechanism: An atom moves by jumping into an adjacent vacant lattice site. The diffusion rate is proportional to the concentration of vacancies.
 - ② Interstitial Mechanism: This is a fast process where an atom (typically a small impurity like H, Li) moves by hopping between the interstitial voids in the crystal lattice.
 - ③ Interstitial-based Mechanisms: Many dopants (like B, P in Si, or Zn in GaAs) are too large to be simple interstitials but diffuse faster than the vacancy mechanism alone would allow. They use complex pathways. The "kick-out" mechanism is where a fast-diffusing interstitial impurity kicks a host atom off its site and takes its place.

Characterization Techniques

- Structural technique: Transmission Electron Microscopy (TEM) is the standard for visualizing defects. By transmitting electrons through ultra-thin samples, it provides atomic-resolution images of physical flaws like dislocations, grain boundaries, and precipitates.
- Electrical technique: Deep Level Transient Spectroscopy (DLTS) is used for quantifying deep-level traps. By measuring how capacitance changes as defects thermally release trapped carriers, it generates a fingerprint of the defect's energy level and concentration.
- Optical technique: Photoluminescence (PL) is a non-destructive test for material quality. It analyzes light emitted by the crystal after laser excitation. Sharp peaks identify specific impurities, while low light intensity indicates a high density of detrimental defects, like non-radiative recombination centers, which introduce a localized energy state deep within the semiconductor band gap, acting as an intermediate catalyst for the annihilation of an electron-hole pair.

Quantum Defects

- Classical point defects (like Boron or Phosphorus in Silicon) create energy levels very close to the Conduction or Valence bands. Thermal energy is enough to ionize them, releasing electrons/holes into the band to conduct electricity.
- Quantum defects create energy levels deep within the bandgap, trapping the electron at the defect site thus making it highly localized. Because the electron is trapped, it behaves like a single isolated atom floating in a vacuum and retains discrete energy levels and quantum spin states.
- For example, the Nitrogen-Vacancy (NV) Center in diamond is a quantum defect, where the nitrogen atom replaces a carbon atom and the adjacent spot is empty. It acts like a trapped ion frozen in the diamond lattice. Lasers can be used to read its spin state, and microwaves can be used to flip its spin.

Applications of NV Quantum Defects - Qubits and Quantum Sensors

- Diamond has extreme structural rigidity which minimizes the lattice vibrations (phonons). This provides a vacuum-like environment that prevents thermal-noise from disrupting electron spin, thus preserving quantum coherence.
- In the absence of a magnetic field, the $|+1\rangle$ and $|-1\rangle$ states are degenerate. When a magnetic field is applied, the Zeeman Effect splits them where the $|+1\rangle$ state moves up in energy and the $|-1\rangle$ state moves down in energy. This energy is directly proportional to the magnetic field strength (B).
- 2.87 GHz is the natural energy gap between the qubit's ground state ($|0\rangle$) and the excited states ($|\pm 1\rangle$) when no external magnetic field is present.

Applications of NV Quantum Defects - Qubits and Quantum Sensors

NV Centers in Diamond as Qubits:

- 1 The electron in NV center can be in either $|0\rangle$ or $|\pm 1\rangle$. If it is in the $|\pm 1\rangle$ states, it de-excites through a non-radiative path that forces it to fall back into the $|0\rangle$ state without emitting a photon. If the electron is in the $|0\rangle$ spin state, it absorbs the light and emits red light and falls back to $|0\rangle$, effectively polarizing the spin system into $|0\rangle$ state.
- 2 By applying a precise pulse of microwaves we can flip the spin from $|0\rangle$ to $|1\rangle$. This is how we perform logic gates.
- 3 In Spin-Dependent Fluorescence we shine Green Laser. If the red-light emitted is bright = $|0\rangle$, and if it is dim = $|\pm 1\rangle$. By counting photons with a photodiode, we can determine the qubit's state.

Applications of NV Quantum Defects - Qubits and Quantum Sensors

NV Centers in Diamond as Quantum Sensors (Optically Detected Magnetic Resonance):

- 1 To measure the field, we don't measure the energy directly, we measure the microwave frequency that flips the spin. The system operates by maintaining continuous green laser illumination to keep the optical readout active, while simultaneously sweeping the microwave drive frequency across the relevant spectrum.
- 2 When the microwave frequency precisely matches the energy required for the $|0\rangle \rightarrow |-1\rangle$ transition, the spin flips to the $|-1\rangle$ state. This results in a measurable dip in fluorescence intensity because the $|-1\rangle$ state is less bright than the $|0\rangle$ state.
- 3 The magnitude of the external magnetic field is calculated by observing how far the resonance frequency shifts from 2.87 GHz. This shift is converted into magnetic field strength using the gyromagnetic ratio ($\gamma \approx 28 \text{ MHz/mT}$).

Thank You!